Linear Algebra Assignment -2

① let there exists two distinct matrices B and B' such that both B and B' are left inverses of A.

$\Rightarrow BA = I$, $B'A = I$

$\Rightarrow$ Now, let C be the right inverse of A.

$\Rightarrow$ we know that for a $n \times n$ ~~matrices~~ matrix, its left inverse is same as right inverse

i.e. $B = C$, $B' = C$

Proof: $B = BI$

$= B(AC)$

$= (BA)C$

$= I \times C$

$\therefore \boxed{B = C}$, similarly $B' = C$.

$\Rightarrow B = IB$

$= (B'A)B = (B'A)C$ $\quad \because (B = C)$

$= B'(AC)$

$= B' \times I$.

$\therefore \boxed{B = B'}$

Hence it is impossible for a $n \times n$ matrix to have two left inverses, $\exists$ only one unique left inverse

② For a system of non-homogeneous linear equation $AX = Y$.

the augmented matrix A' is

$A' = [A \mid Y]$

Method- 1

In Ex-15

using elementary row operations the augmented matrix is reduced until the left side becomes the Identity matrix I.

If A is invertible, row reduction transforms this matrix into $[I \mid Z]$

Z is the solution of X.

$$X = Z = A^{-1}Y.$$

x can be written as linear function of Y.
To obtain A^{-1}, choose Y to be the standard basic vectors

$e_1, e_2, \ldots\ldots e_n.$

For each i,

$AX = e_i$

Solved by row, reducing A.

$[A \mid e_i]$

let the sol^n be x_i

The vectors form the column A^{-1}, i.e

$A^{-1} = [x_1, x_2, x_3, \ldots, x_n]$

$$[A \mid Y] = \begin{bmatrix} 2 & -1 & y_1 \\ 1 & 3 & y_2 \end{bmatrix}$$

$R_2 \leftrightarrow R_1$

$$[A \mid y] = \begin{bmatrix} 1 & 3 & y_2 \\ 2 & -1 & y_1 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$

$$[A \mid y] = \begin{bmatrix} 1 & 3 & y_2 \\ 0 & -7 & y_1 - 2y_2 \end{bmatrix}$$

$R_2 \to -R_2/7$

$$\begin{bmatrix} 1 & 3 & y_2 \\ 0 & 1 & \frac{-1}{7}(y_1 - 2y_2) \end{bmatrix}$$

$R_1 \to R_1 - 3R_2$

$$\begin{bmatrix} 1 & 0 & \frac{1}{7}[y_2 + 3y_1] \\ 0 & 1 & \frac{1}{7}(2y_2 - y_1) \end{bmatrix} ; \quad e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$y_1 = 1 ; \; y_2 = 0$

$$x_1 = \begin{bmatrix} 3/7 \\ -1/7 \end{bmatrix} \Rightarrow y_1 = 0, \; y_2 = 1 \Rightarrow e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 1/7 \\ 2/7 \end{bmatrix} ; \text{ so, } A^{-1} = [x_1 \; x_2] = \begin{bmatrix} 3/7 & 1/7 \\ -1/7 & 2/7 \end{bmatrix}$$

<u>Method - 2</u>

Instead of solving $AX = Y$ Seperately for each basis vector we form the augmented matrix.

$$[A \mid I]$$

Apply elementary row operations to reduce A to I and the same operations on I, we get

$$[I \mid A^{-1}]$$

This works because, row operations correspond to left multiplication by invertible matrices

<u>Ex: 16</u>

$$A = \begin{bmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{bmatrix}.$$

$$A \cdot A^{-1} = I$$

$$\begin{bmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

After row-reducing

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A^{-1} = \underbrace{\begin{bmatrix} 9 & -36 & 30 \\ -36 & 192 & -180 \\ 30 & -180 & 180 \end{bmatrix}}_{A^{-1}}$$

be zero, Hence $R = I$.

(3) To solve $AX = Y$, we form the augmented matrix $A' = [A \mid Y]$, Now, we perform elementary row operations on A', to get R, which is the RREF of A.

$\Rightarrow RX = Z$ has the same sol^n as that of $AX = Y$

$\Rightarrow A' = [A \mid Y] = [R \mid Z]$

$\Rightarrow$ For soln to exist, for all zero rows of R, corresponding entry in z must also be 0.

Given $A = \begin{bmatrix} 0 & 5 & -1 \\ 1 & 8 & -1 \\ 2 & 1 & 1 \end{bmatrix}$

$$\Rightarrow A' = \begin{bmatrix} 0 & 5 & -1 & y_1 \\ 1 & 8 & -1 & y_2 \\ 2 & 1 & 1 & y_3 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} 1 & 8 & -1 & y_2 \\ 0 & 5 & -1 & y_1 \\ 2 & 1 & 1 & y_3 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 + (-2)R_1} \begin{bmatrix} 1 & 8 & -1 & y_2 \\ 0 & 5 & -1 & y_1 \\ 0 & -15 & 3 & y_3 - 2y_2 \end{bmatrix}$$

$$\xrightarrow{R_2 \to R_2/5} \begin{bmatrix} 1 & 8 & -1 & y_2 \\ 0 & 1 & -1/5 & y_1/5 \\ 0 & -15 & 3 & y_3 - 2y_2 \end{bmatrix}$$

$$\xrightarrow[R_3 \to R_3 + (15)R_2]{R_1 \to R_1 + (8)R_2} \begin{bmatrix} 1 & 0 & 3/5 & y_2 - 8y_1/5 \\ 0 & 1 & -1/5 & y_1/5 \\ 0 & 0 & 0 & y_3 - 2y_2 + \frac{7}{5}y_1 \end{bmatrix}$$

Hence for solutions to exist,

$$\boxed{y_3 + 3y_1 - 2y_2 = 0}$$

Also, $x_1 + \frac{3x_3}{5} = y_2 - \frac{8y_1}{5}$

$x_2 - \frac{x_3}{5} = \frac{y_1}{5}$

$\because x_3$ is a free variable, it can be taken as some $c \in F$.

$$\boxed{\begin{aligned} \therefore\ x_1 &= \frac{-3c}{5} + y_2 - \frac{8y_1}{5} \\ x_2 &= \frac{c}{5} + \frac{y_1}{5} \ ; \ x_3 = c \end{aligned}}$$

4) If A is invertible, then

$AX = Y$.

$X = A^{-1}Y$ is a solution

$AX = Y$

$A^{-1}AX = A^{-1}Y$

$X = A^{-1}Y$

For any vector Y, $A^{-1}Y$ exist

So X will always exist

(i) $\Rightarrow$ (ii)

Let's take

$AX = Y$ has sol^n for every Y.

let R be row reduced echelon matrix which is equivalent to A.

we will show $R = I$

To show that a row-reduced ~~mat~~ matrix R is identity we just need to show that the last row of R is not all zeroes.

let $E = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Claim: $RX = E$ can be solved for X.

we know $R = PA$.

where P is invertible.

Thus $RX = E$

$\Rightarrow PAX = E$

$AX = P^{-1}E$

Since $AX = Y$ has a solution $P^{-1}E$ exists.

the rightmost system has a solution. Thus, x also solves $RX = E$, since this can be solved for x then the last row of R cannot be zero

Hence, $R = I$.

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(5) $\because$ A is an invertible square matrix, let
$AA^{-1} = I = A^{-1}A$
$\Rightarrow$ Given that $A = A_1 A_2 \cdots A_k$.
$\Rightarrow A_k(A_1 A_2 A_3 \cdots A_{k-1}) = I$.
$\therefore$ A_k is also invertible with $A_k^{-1} = A^{-1} A_1 A_2 \cdots A_{k-1}$
Similarly $A_{k-1}(A_k A^{-1} A_1 \cdots A_{k-2}) = I$.
Hence A_{k-1} is also invertible.
$\therefore$ continuing the same for $A_{k-2}, A_{k-3}, \ldots A_1$
Hence, if A is invertible, A_j is also invertible.

(6) Every elementary row operation can be written as multiplication by an elementary matrix.
~~Exact~~ Each elementary row operation can be written as multiplication by an elementary matrix, so the sequence row operation is represented by matrices $E_1, E_2, \ldots E_k$.
This operations reduce A to identity matrix.

~~$E_k E_{k-1} E_k$~~ $E_k E_{k-1} E_{k-2} \cdots E_1 A = I$

Because A is invertible,
multiply with A^{-1} on both sides

$E_k E_{k-1} E_{k-2} \cdots E_1 I = A^{-1}$ — (1)

Now take identity matrix & apply the same sequence of row operations to identity matrix I.

$E_k E_{k-1} E_{k-2} \cdots E_1 I = A^{-1}$
From eqn (1)
Apply the same row operations to I gives A^{-1}.

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(7) let the left inverse of A be B.

$\Rightarrow BA = I$.

Given $AX = 0$.

Multiplying B on both sides,

$BAX = B \times 0 = 0$

But $BA = I$.

$\Rightarrow I \times X = 0$

$\therefore \boxed{X = 0}$

$\therefore AX = 0$ has only trivial solution.

Hence proved.

(8) let A be a $m \times n$ matrix over a field F.

let $u_1, u_2, u_3, \ldots, u_r$ be the non-zero rows of the row-reduced echelon form of A [RREF].

In RREF, each non-zero row has a leading 1 (pivot).

Except pivot all other entries in that column are zero.

The pivot positions of each row is ~~at~~ in a different column.

So, no row can be written as a combinⁿ of other rows.

Hence $u_1, u_2, \ldots, u_r$ are linearly dependent.

$\therefore c_1 u_1 + c_2 u_2 + \cdots + c_r u_r = 0$.

$\Rightarrow c_1 = c_2 = c_3 = \cdots = c_r = 0$.

This statement is true. The non-zero rows of the RREF are linearly independent, so all coefficients must be zero.